NVIDIA Nemotron 3 Nano Omni: One Open Model for Vision, Audio, and Text at 9x the Throughput

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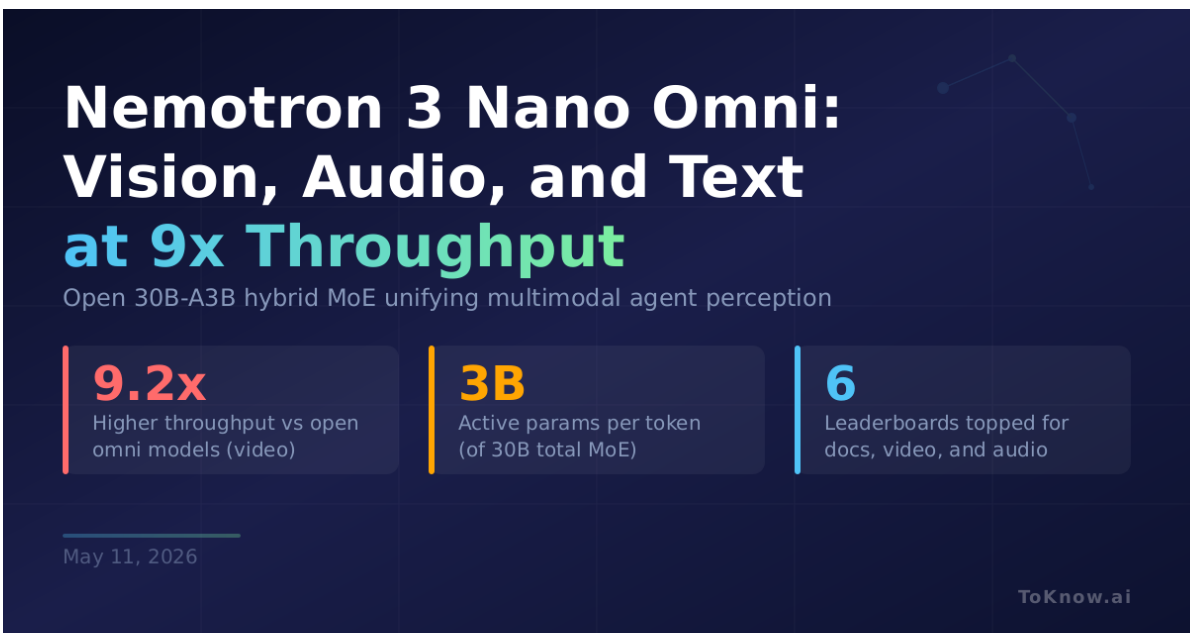

NVIDIA released Nemotron 3 Nano Omni on April 28, a 30-billion-parameter open multimodal model that handles text, images, audio, and video in one architecture. The "30B-A3B" label means 30 billion total parameters but only about 3 billion active per token, thanks to a hybrid Mixture-of-Experts design that blends Mamba layers (for memory-efficient sequence processing) with transformer layers (for precise reasoning). NVIDIA reports 9.2x higher system throughput for video tasks and 7.4x for multi-document tasks compared to other open omni models at the same per-user responsiveness threshold. It tops six leaderboards for document intelligence, video understanding, and audio comprehension, including OCRBenchV2 and VoiceBench. The model ships with open weights, training data, and full recipes, and runs on everything from Jetson edge hardware to cloud GPUs via Ollama, vLLM, and 25+ partner platforms.

The payoff is collapsing three or four separate models (vision, speech, language, document parsing) into one. An AI agent that needs to watch a screen recording, read a chart, and listen to a voice note can now do it in a single inference loop instead of chaining separate models. H Company's computer-use agent, built on Nano Omni, processes full 1920x1080 screen recordings in real time. Since only 3B parameters fire per token, the model fits on hardware that a dense 30B model never could.

Multimodal AI is moving from "bolt separate encoders onto a language model" to unified architectures where vision, audio, and text share the same reasoning loop. LongCat-Next showed a similar approach at 74B total (3B active), and now NVIDIA's entry brings full-stack deployment support from edge to cloud.

Sources:

- Nemotron 3 Nano Omni Technical Report (arXiv)
- NVIDIA Technical Blog: Nemotron 3 Nano Omni
- NVIDIA Blog: Launch Announcement
- Nemotron 3 Nano Omni on Hugging Face
